Physical Chemistry CUME, Jan. 25th, 2019

Statistical Thermo-dynamics

The number of ways to distribute *N* particles between two boxes satisfies the binomial distribution: $W = \frac{N!}{M! \cdot (N-M)!}$, where *M* is the number of particles in the first box and *N-M* is the number of particles in the second box.

The entropy of a system can be expressed as $S = k_B \ln(W)$, where k_B is the Boltzmann's constant. And for a large number, $\ln(N!) = N \cdot \ln(N) - N$.

(a) Consider a system that contains two boxes of equal size in close contact. Initially, the left-side box is filled with 1 mole (N_A) of ideal gas, and the right side box is empty. A hole is poked to allow the ideal gas from the left-side box to expand isothermally to the right-side box. Please work out the entropy change for this process. (Hint, you can use the information given in this CUME about binomial distribution to solve this problem. It will be fine if you use what you learnt in your pchem class to solve this. The result should be the same.)

(b) Consider a system that contains two boxes of equal size in close contact. Initially, the left-side box is filled with 1 mole of Ar and the right side box contains 1 mole of Ne. A hole is poked to allow gases to mix. Assume the mixing process is isothermal. Please calculate the entropy change associated with mixing. (Hint: similarly you can use the binomial distribution or any other method appropriate for this problem.)

(c), Assume 6 pchem students are enrolled in the Department of Chemistry and Biochemistry graduate program in Fall 2019, how many ways are there for the 6 students to join 3 different pchem groups (In reality we have more than 3 groups, this is just to make your math easier) with 2 students in each group.